Bridging Neural and Behavioural Data with XAI in Predictive Neuromarketing

Final Year Research Project — Department of Computer Science, Sri Lanka Institute of Information Technology

Group No: 25-26J-145-DS

PP2 Comments & Actions Taken

Comment	Action Taken
Expected to see EEG signals — collect through real participants	EEG data collection protocol designed; pilot recordings scheduled with consenting participants using 32-channel EEG. Preprocessing pipeline implemented.
Improve UI	UI refinements completed: dashboard layout updated, clearer visual hierarchy, accessible colour contrast.
Present XAI as separate modules if integration not possible	XAI developed as modular components (Neural, Behavioural) and also demonstrated in an integrated pipeline.
Use more federated learning techniques	Federated averaging and client-side regularization added; experiments run with simulated edge clients.

These actions address reviewer feedback while preserving experimental integrity and participant privacy.

Research Problem

Existing neuromarketing systems often predict consumer decisions without clearly explaining the reasoning behind them. They rely on costly hardware, offer limited behavioural interpretation, use black-box AI models, and collect data centrally — creating challenges for scalability, trust, ethics, and privacy.

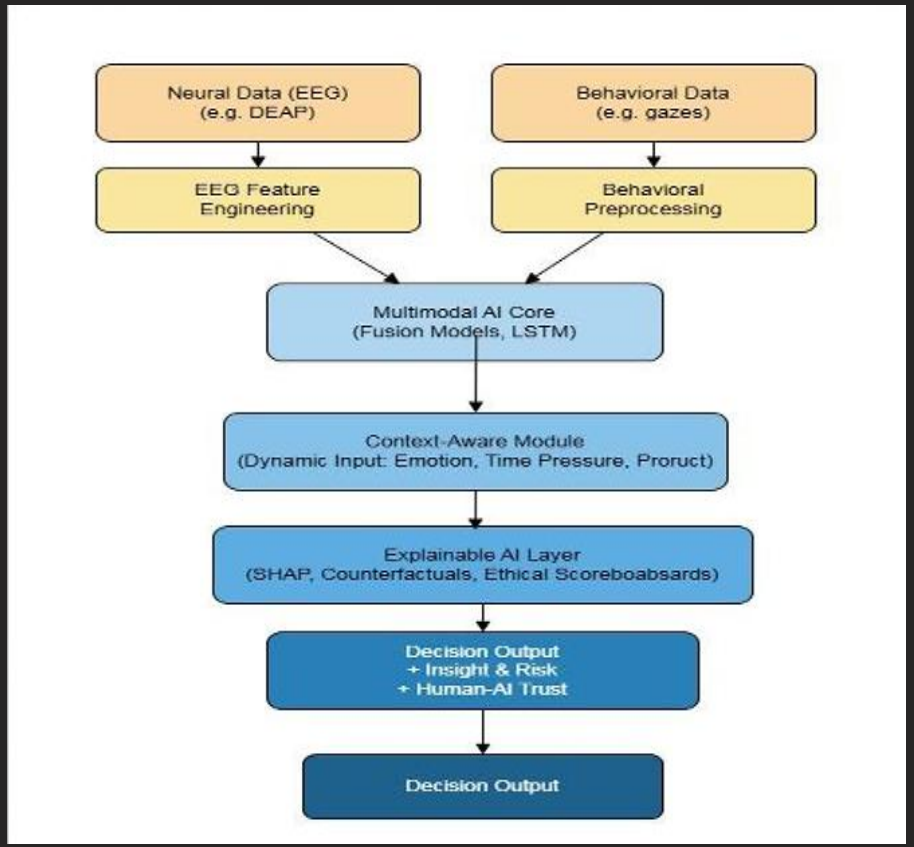
Research question: How can neural and behavioural data be combined to build a predictive neuromarketing framework that accurately forecasts consumer decisions, explains model reasoning with XAI, and preserves user privacy through federated learning?

Key issues addressed

- Limited integration of neural and behavioural data
- Lack of decision-reason prediction
- Black-box AI models with limited ethical transparency
- Privacy risks from centralized data collection
- Need for scalable, practical neuromarketing solutions

Goal: Build an interpretable, privacy-preserving predictive pipeline that fuses EEG and behavioural signals for robust consumer decision prediction.

System Architecture



- **Behavioural Data Module:** PsychoPy experiments capturing gaze, mouse movement, reaction time; lightweight mouse-only fallback for scale.
- **Neural Integration Module:** EEG preprocessing, feature extraction (time/frequency), source-level summaries and synchrony metrics.
- **Explainable AI Module:** SHAP at feature level, counterfactual generators for scenario analysis, dual presentation (local + global explanations).
- **Federated Learning Module:** Client updates, secure aggregation (FedAvg), client-side regularisation and differential privacy options.

Pipeline supports both integrated and modular evaluation paths to satisfy evaluation and deployment constraints.

Design Excellence — How the Proposed Solution Improves Existing Systems

Neural + Behavioural Fusion

Combines EEG features with gaze, reaction time and mouse dynamics for richer signals.

Dual-layer Explainability

SHAP for feature-level attribution and counterfactuals for causal “what-if” insights.

Scalable Tracking

Mouse-based and lightweight behavioural capture to reduce hardware dependence.

Federated Privacy

Decentralised training to keep raw EEG/behavioural data on client devices.

Ethical Metrics

Trust index, bias detection and manipulation risk scoring embedded.

IT22083746 — Ranawakarachchi C P

Component: Neural Integration for Consumer Insights

- EEG acquisition protocol: Multi-channel EEG recordings collected during participant decision-related activities with synchronized activity windows.
- Preprocessing: EDF loading using MNE, bandpass filtering, artifact/noise handling, segmentation based on activity windows, and extraction of decision-related EEG segments.
- Features: Alpha and beta spectral power features, alpha-beta ratio, frontal asymmetry, window-based EEG statistics, and participant familiarity information.
- Models and analysis: Logistic Regression and Random Forest models for decision prediction, along with cognitive workload estimation using regression-based evaluation metrics.
- Output: Prediction probabilities, workload analysis metrics, feature distribution visualizations, and interactive dashboard-based interpretation using React and FastAPI.

IT22150820 — Kavindya T A S

Component: Behavioral Dynamics in Decision Making

Scenarios / Description

Experiment Design

Construct controlled choice tasks and stimuli sequences using PsychoPy.

Behavioural Capture

Record gaze, RT, and mouse dynamics; fallback mouse tracking for wide deployment.

Feature Extraction

Compute fixation counts, saccade metrics, dwell time, hesitation markers and trajectory curvature.

Fusion Prep

Time-align behavioural features with EEG epochs for multimodal modelling.

IT22120984 — Senanayaka I H D M

Component: Ethically Transparent Explainable AI Framework

Explainability components and workflow:

Reports present both human-readable rationales and quantitative explanation scores for stakeholder review.

Global Attribution (SHAP)

Feature importance across cohorts to reveal drivers of decisions.

Counterfactuals

Generate minimal changes that alter predicted outcomes for actionable insights.

Ethical Metrics

Trust index, bias detection and manipulation risk scores accompany predictions.

IT22071316 — Shahaam M M

Component: Contextual Decision Modelling with Federated Learning

- Client setup: local feature extraction for EEG & behaviour; only model updates shared.
- Server aggregation: FedAvg with secure aggregation and optional differential privacy.
- Personalisation: client-side fine-tuning to adapt to individual differences in neural signatures.
- Evaluation: cross-client validation, privacy budget monitoring, communication efficiency experiments.

Federated approach reduces privacy risk while enabling diverse participant data to improve generalisation.

Market Potential & Commercialisation

Target users

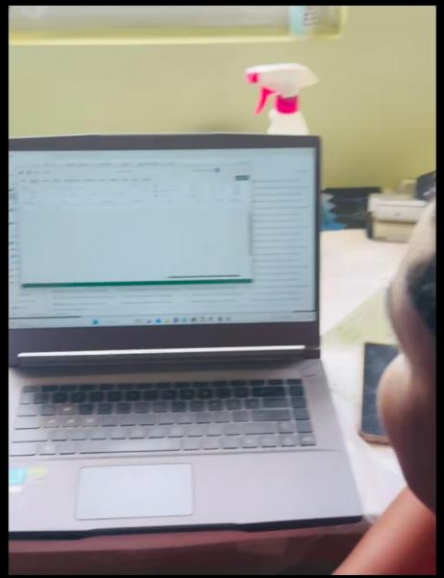
- Marketing agencies seeking evidence-based consumer insights
- Retail & e-commerce brands optimising UX and conversion
- Consumer research institutions and academic labs

Opportunities & Revenue Model

- Productise as SaaS dashboard + research licensing for longitudinal studies
- White-label analytics for agencies, per-seat subscriptions and enterprise licensing
- Consulting and custom experiment design services

Ethics and explainability are differentiators that increase market trust and adoption.

Sleep Disordered Breathi



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27th April 2026

Mr. Chamie Ranawakaarachchi,
Undergraduate,
SLIIT.

Request for Ethical approval of research proposal – ""Bridging neural and behavioral data with XAI in predictive neuro marketing ".

I am pleased to inform you that the Ethics & Research Committee of Lanka Hospitals has granted ethical clearance and permission on 09.04.2026 to conduct the above research.

We wish to inform you that permission granted by the Ethics & Research committee will be valid for one (1) year from the date of approval. In the event your research extends beyond one (1) year you are requested to resubmit your proposal to the committee in order to sustain the approval.

While you conduct your study you are requested to observe,

1. Not to disturb the staff/patient in any manner.
2. To collect only the data that was mentioned in the proposal for which the approval is given.
3. It is prohibited to use data collected for any other purpose other than the study, approved.
4. The name of the research institution (The Lanka Hospitals Corporation PLC) & Patient details **should not be mentioned** in the report.
5. Other than the submission you make for the purpose of your research you should not publish the data collected through the study in any other publication.
6. A Copy of the report prepared after the study should be given to Lanka Hospitals.

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Dr. Sanjaya Ratnayake
Group Director Medical Services
Chairman - The Lanka Hospitals Ethics & Research Committee

Organization Accredited by Joint Commission International
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Thank You!